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Conclusion

Software architectures

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Outline

1. (Quick and dirty) summary of the course
2. Exam
3. Career paths
4. Stage and theses
5. Possible collaborations

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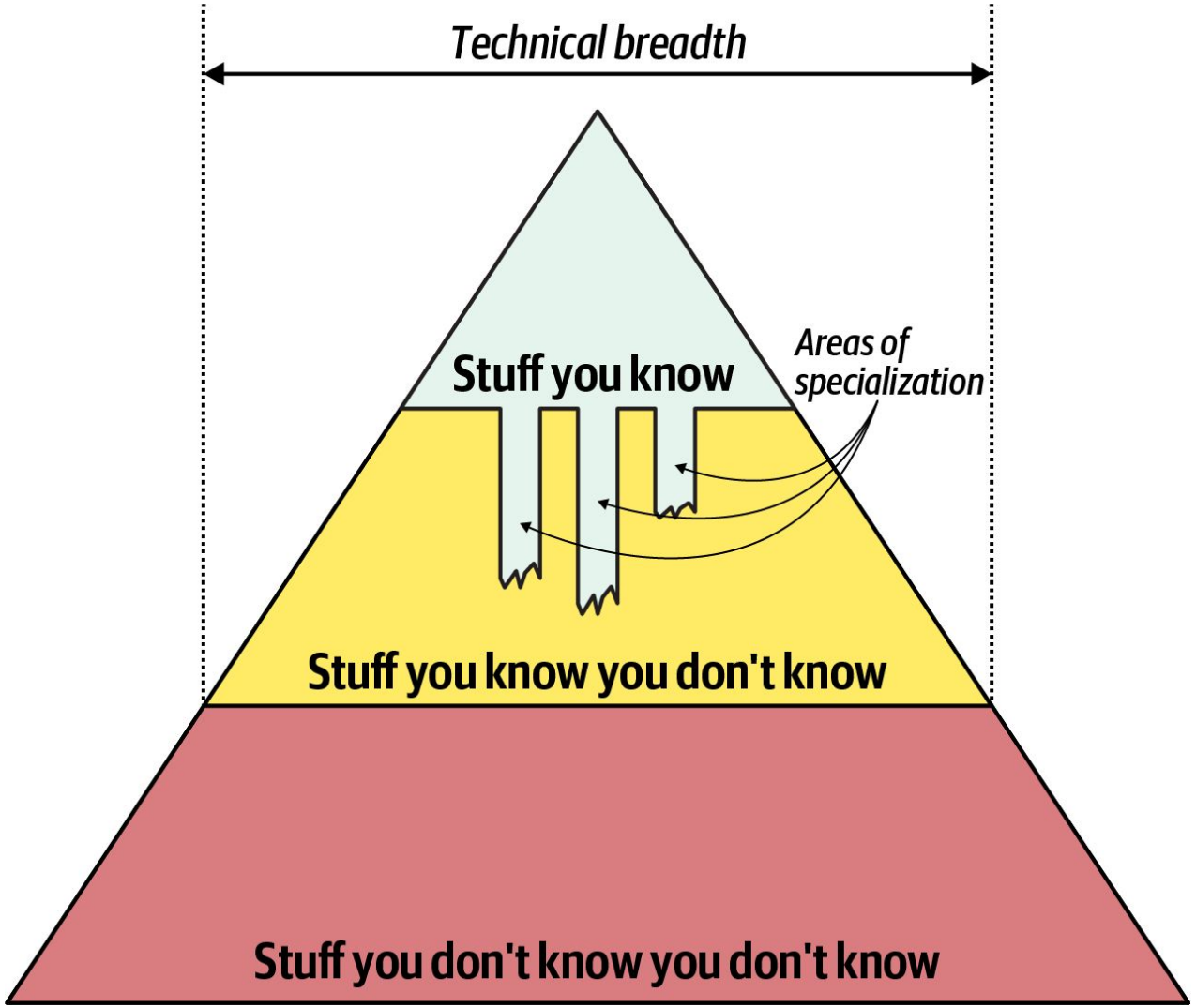
Technical depth and breadth



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Architecture characteristics

Deployability	the ease, frequency, and risk of deployment	**Performance**	amount of used resources, including time behavior, resource utilization, and capacity
Elasticity	the ability to handle bursts of users	**Reliability**	degree to which a system functions under specified conditions for a specified period of time
Evolutionary	the ability accept incremental and guided changes	**Scalability**	ability for the system to perform and operate as the number of users or requests increases
Fault tolerance	does the software operate as intended despite hardware or software faults	**Simplicity**	the ease of understanding the overall system
Modularity	degree to which the software is composed of discrete components	**Testability**	the ease of and completeness of testing
Overall cost	"additional" effort required to implement the system		

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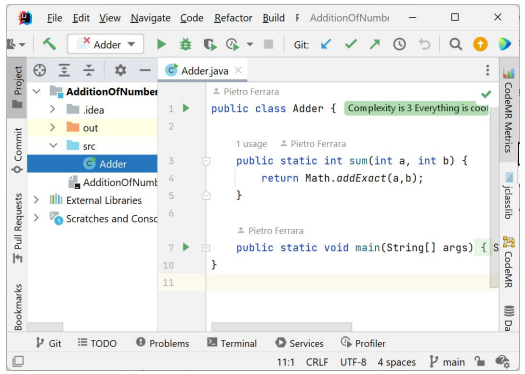
Technologies

Source code



spring

http://localhost:8080

Compile and build

Gradle

spring

Runnable artifact

JAR

Deploy

docker

Server

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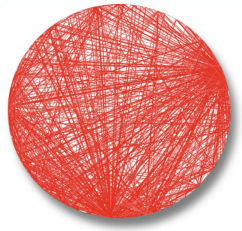
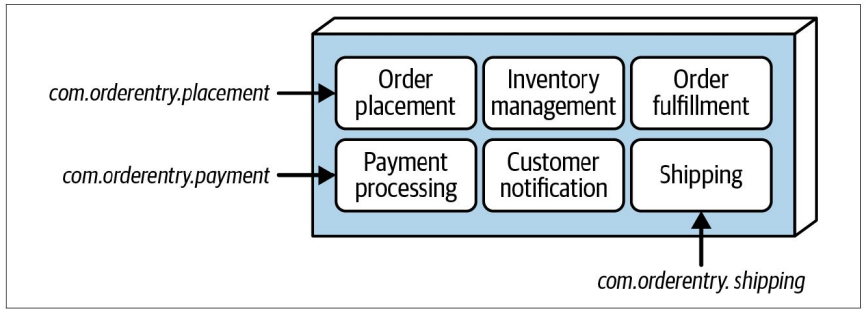
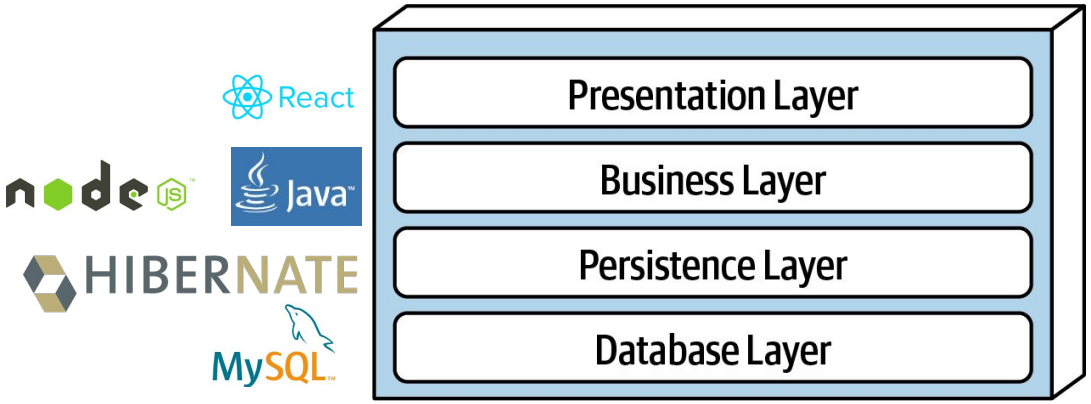
Monolithic architectures



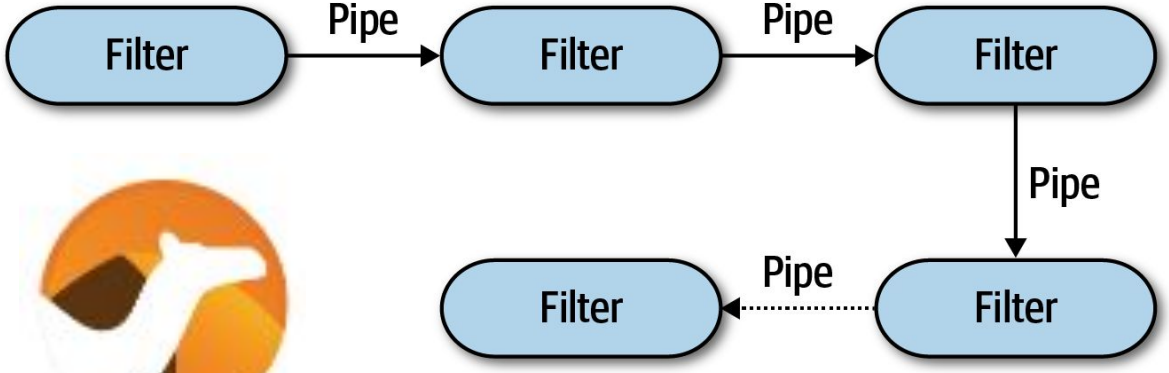
Ball of Mud



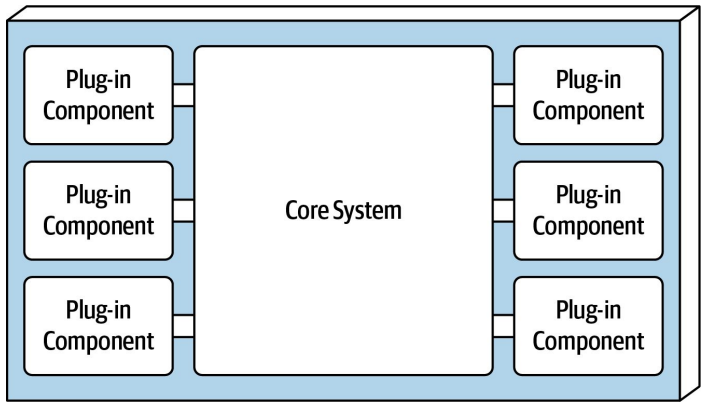
Modular monolith



Layered/ntier



Pipeline



Microkernel/plugin

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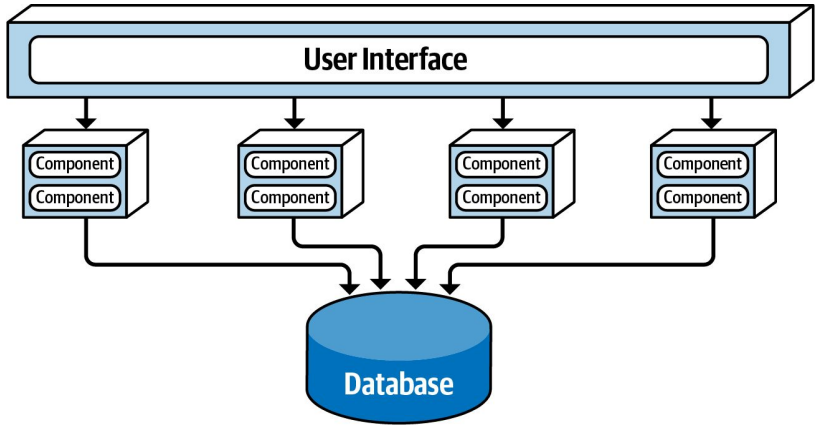
Distributed architectures

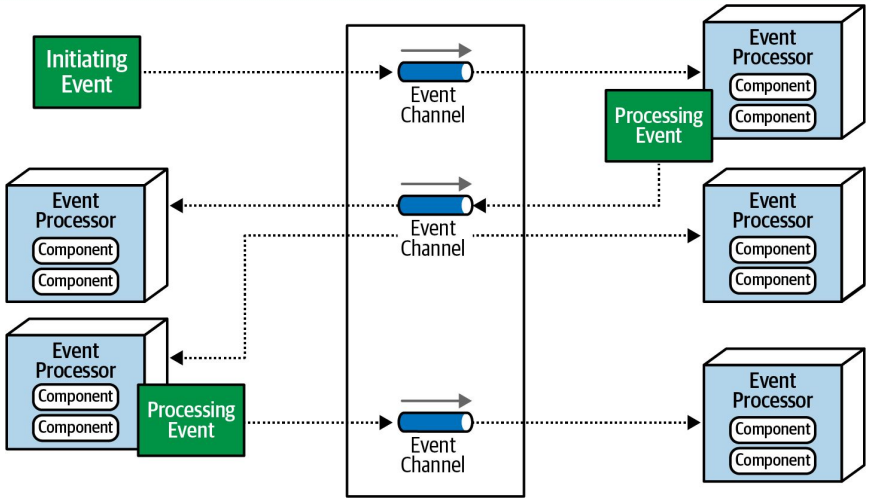


Service-based

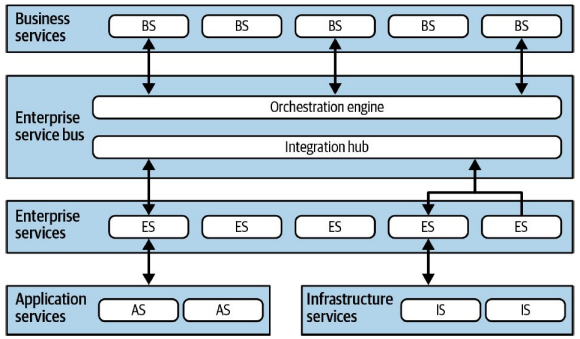
HTTP

REST API

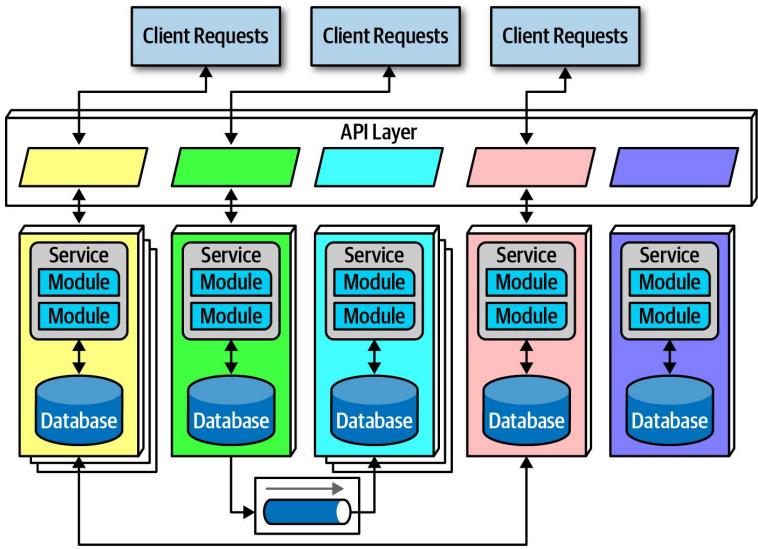
Docker compose



Event-based RabbitMQ



Orchestration-driven



Microservices

kubernetes

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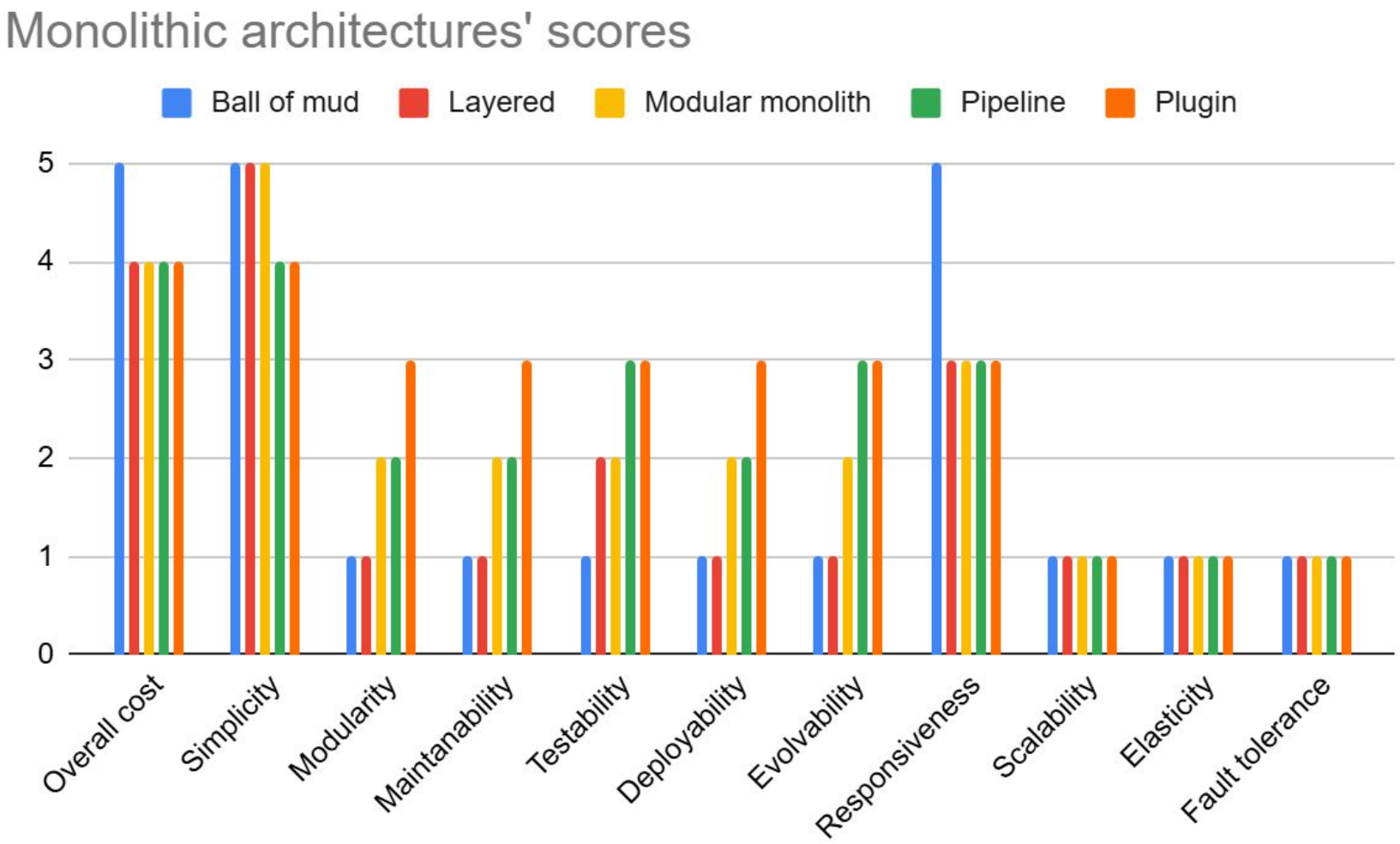
Monolithic architectures' scores



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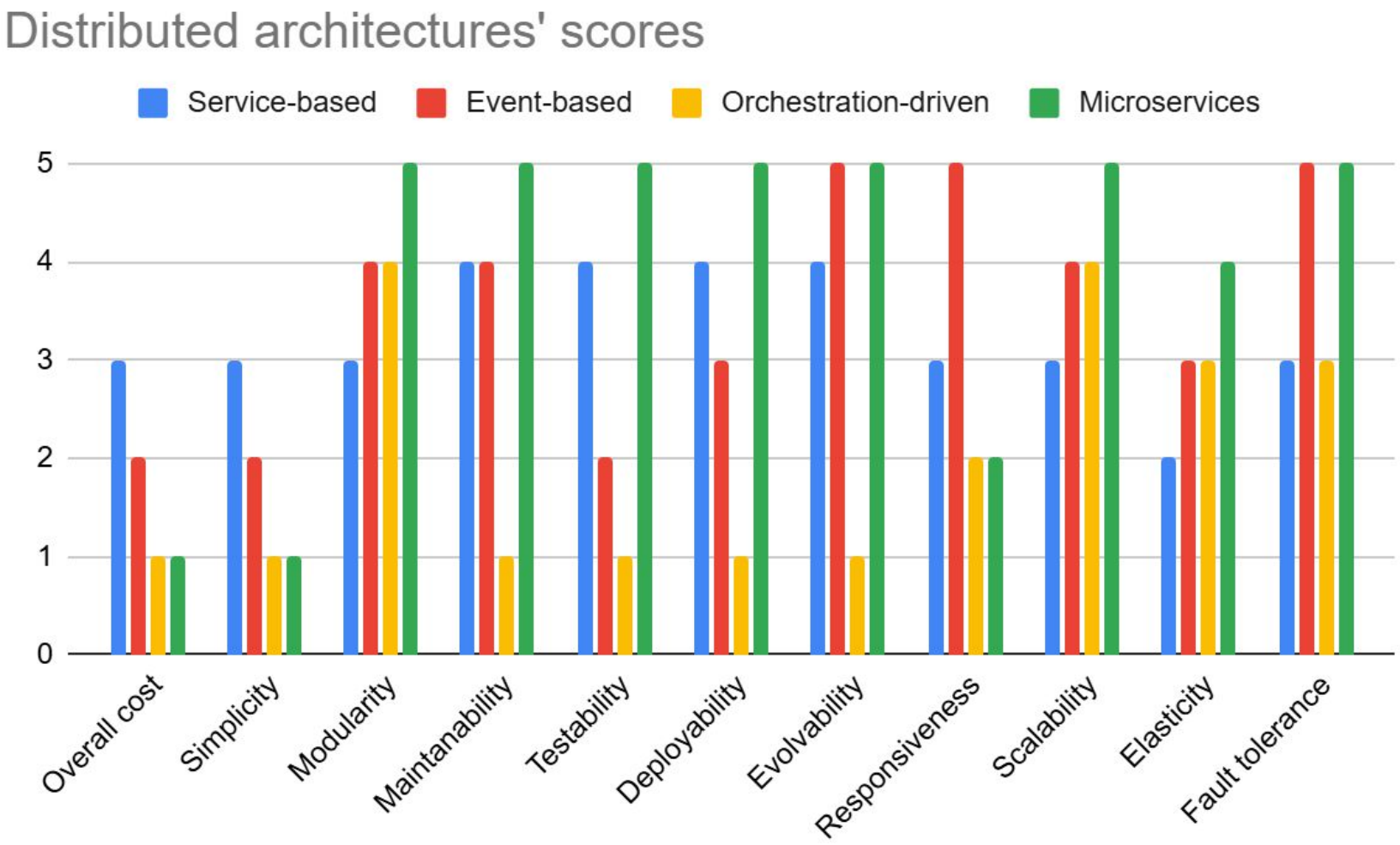
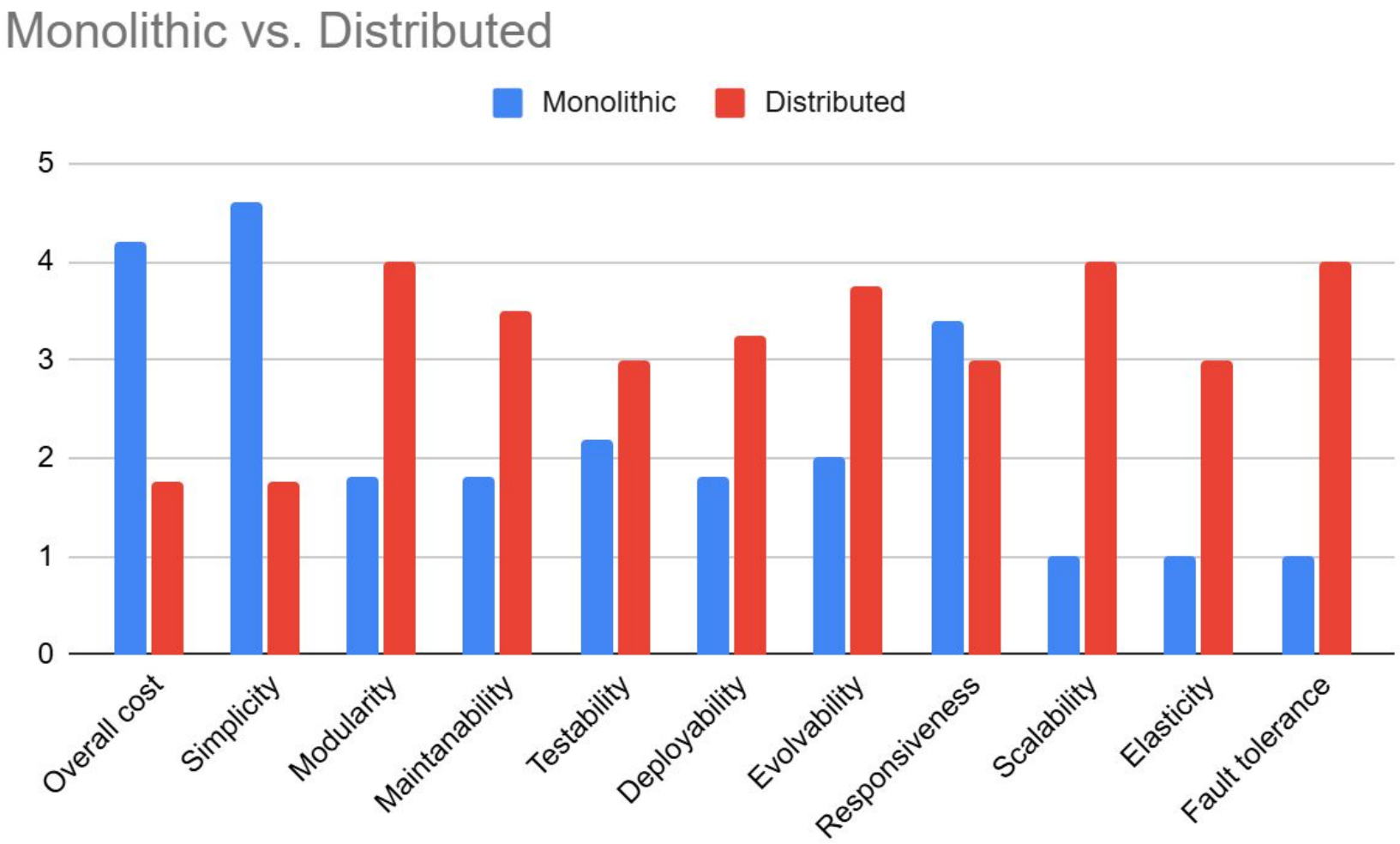
Distributed architectures' scores



Monolithic vs distributed



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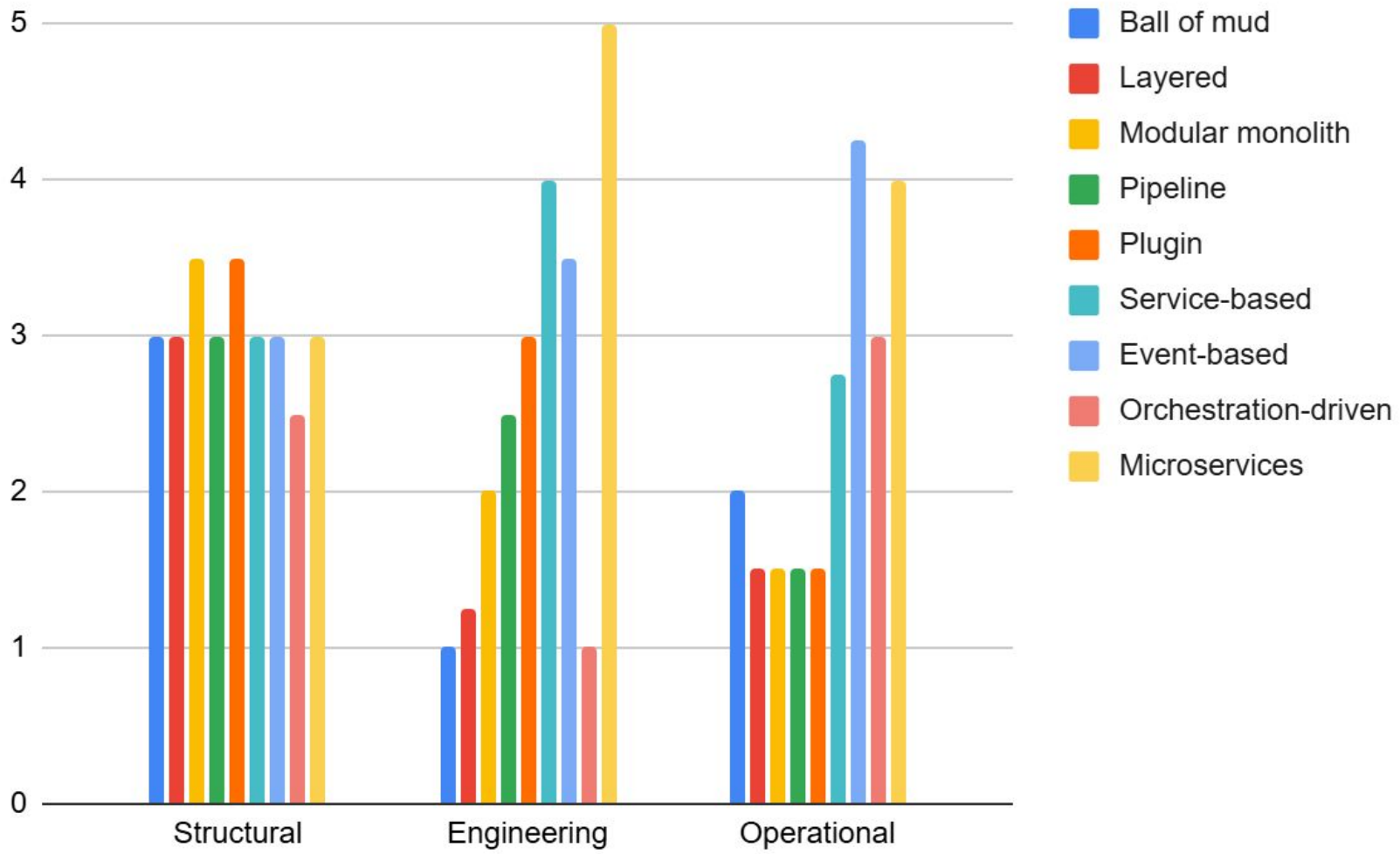
Summarizing characteristics



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The laws of software architecture (Chapter 27)

1. Everything in Software Architecture Is a Trade-Off
   a. Shared Library Versus Shared Service
   b. Synchronous Versus Asynchronous Messaging
2. Why Is More Important Than How
   a. Out of Context Antipattern
3. Most architecture decisions aren't binary but rather exist on a spectrum

Project and written exam (1/2)

- 3 group tasks during the course (two thirds of the grade)
  - 2 small ones at the beginning: DONE!
  - 1 big task at the end implementing the architecture
    - Using **any technology**, not only the ones I'll introduce you
    - Containing **at least one monolithic and one distributed architecture**
    - Deadline: before the written exam (**February 4th**)!
- Written exam at the end of the course (one third of the grade)
  - **Only after submitting and passing** the tasks/project
  - Example from last year in Moodle
  - About **only all** the in presence lectures

Project and written exam (2/2)

- You will have to deliver at the end of the project
  - A GitHub repository containing the **code**
  - The **history of the repo** should make clear the contribution of the different students in the group
  - It must contain a **description** (pdf, markdown, …) of the structure of the architecture with references to the code and **how to run it**
    - Docker (file or compose) or Kubernetes (minikube)

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Example of a possible written exam

- The written exam will be 45 minutes long
- It will consist of 4 questions with open answers
- Each question will count for 2.5 points
- Exams will take place on:
  - February 4th at 8:30am
  - February 18th at 8:30am
- An example of last year exam is published in Moodle
  - Together with a master solution
    - It explains how I would have given points to (aka graded) your answers
  - Note that all the part of invited lectures last year was missing

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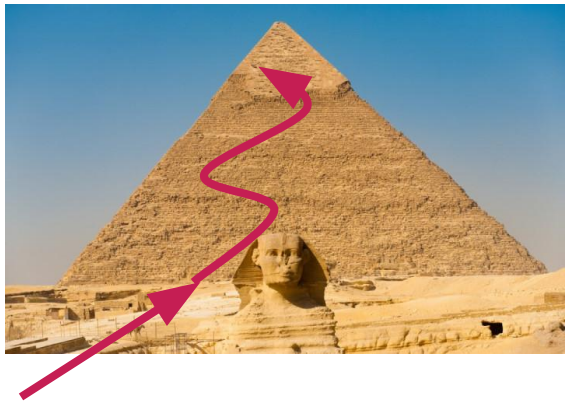
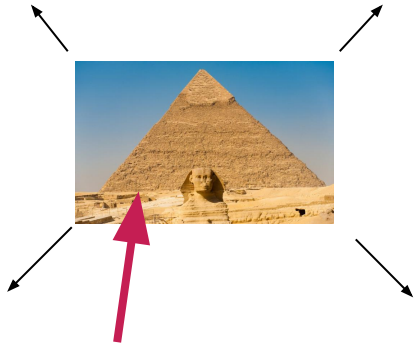
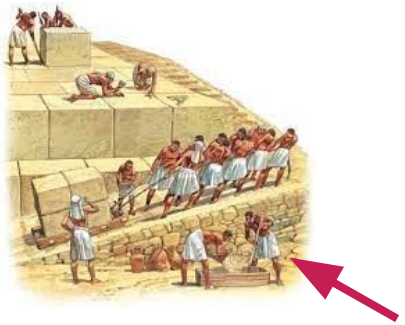
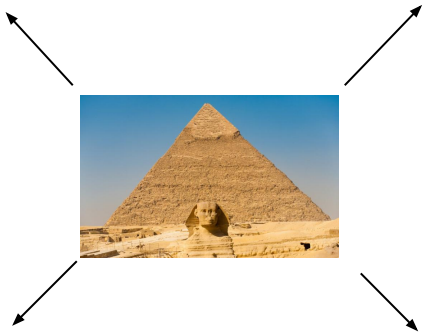
Career paths

Academic	PhD → Postdoc → Professor 1, 2, 3…	Adv.: complete freedom, work certain for life, students Dis.: highly uncertain and competitive, low salary (especially at the beginning)
Employee	 	Adv.: work mostly certain, salary sure at the end of the month Dis.: not so much freedom, "medium" salary
Entrepreneur	 	Adv.: relative freedom (up to the market), potentially a lot of money Dis.: highly uncertain, no sure salary (usually no salary at all for the first periods)

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My career path

Academic

PhD → Postdoc

Thesis about Java multithreading

PhD in CS

Master in CS

ÉCOLE POLYTECHNIQUE

OO programming and Scala

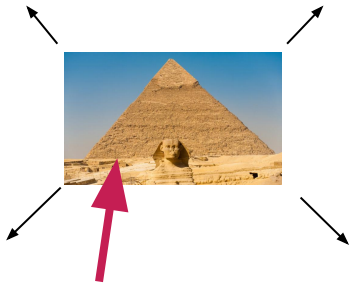
Postdoc and lecturer

ETH zürich

Android and JavaScript

Research Staff Member

IBM



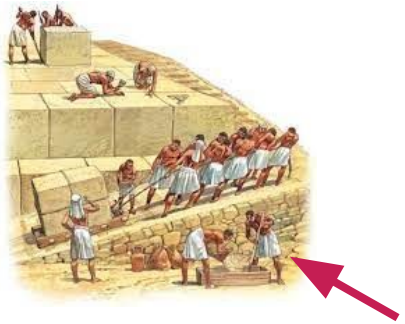
.NET and Java

Head of R&D

JULIASOFT code analysis reinvented

Academic

Professor 1, 2, 3…



Assistant professor

Associate professor

2005 2009 2013 2016 2020 2023

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Main takeaways

1. Choose your initial career path
   a. DON'T do the first thing it comes into your hands!!!
   b. Remember: you will be an extremely strong professional
2. Never think you are stuck in a path
   a. Just change path whenever you think it's worth to

I'm personally open in helping/supervising/collaborating with any possible path, from PhDs to startuppers

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Stage and Thesis

- My main theoretical research interests concern what is taught by prof. Cortesi at Software Correctness, Security and Reliability
  - https://www.unive.it/data/course/354811/programma
  - You'll probably follow it next semester
- My main practical/industrial research interests are mostly focused on software development, engineering, architecture
- Some topic proposals here
- Open to anything else
  - Both for internships or stages with enterprises (my spinoff, other of my contacts, or your contacts)

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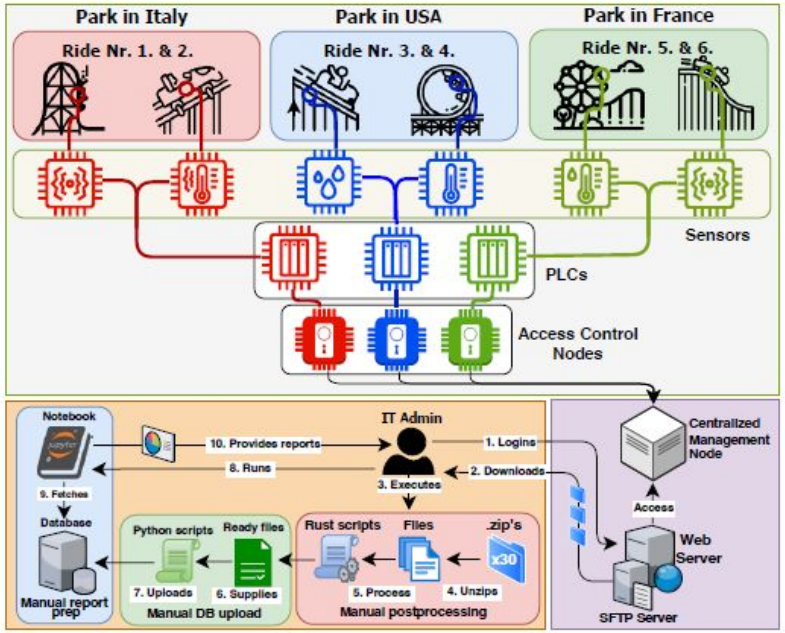
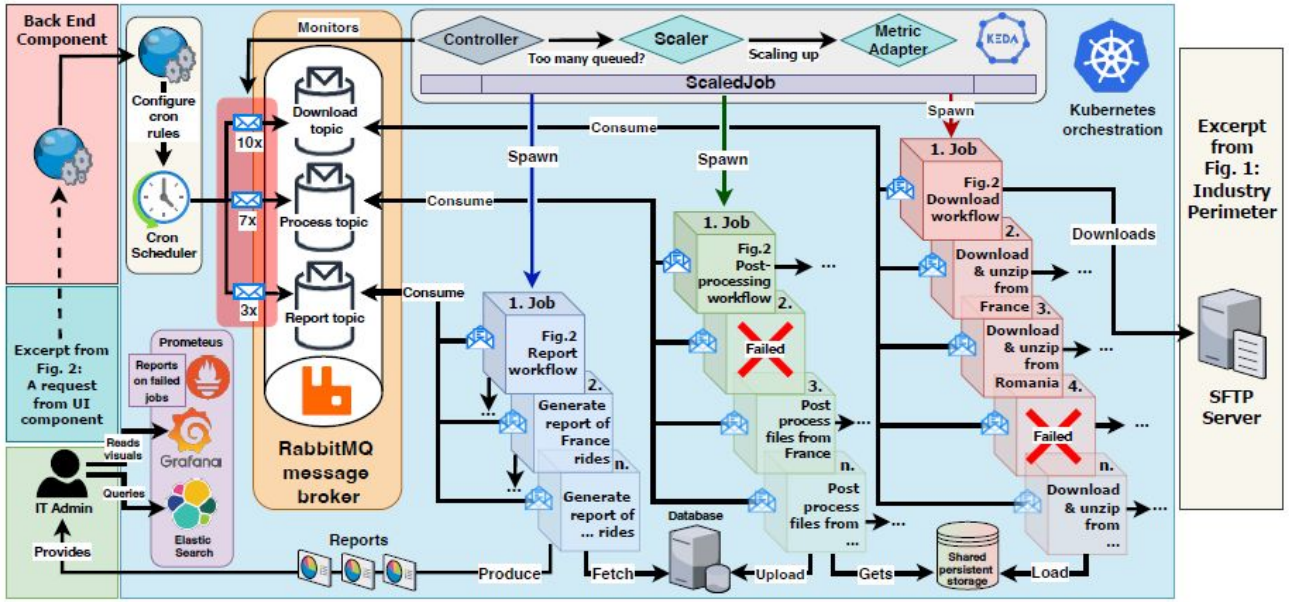
Zamperla

- Project of our spinoff and <u>Zamperla</u>
  - Amusement parks
  - Managing Coney Island in NY
  - Rather big specialized enterprise
- Possible types of collaborations
  - Stage in Zamperla/spinoff
  - Theses
  - Paid consultancy
- Stop/ping me if interested!